

CS F212 DATABASE SYSTEMS 2024-25

LAB 5 EXERCISE (JOINS AND CROSSPRODUCT)

Instructions: The solution can be found in multiple ways. Today's lab is based on Joins and Cross product concepts. Thus solutions using SET operations or nested queries, though correct, will not be considered for evaluation.

Q1 to Q3 and Q5 to Q7 carries 1 mark each
Q4 and Q8 carries 2 marks each.

Solve Q1 to Q4 using Joins concept. No marks will be given if solved using SET operations, subqueries or Crossproduct.

Q1. For each product line, display the products that have been ordered at least once. Display in decreasing order according to the number of ordered products

ProductLine	OrderedProducts
Classic Cars	37
Vintage Cars	24
Motorcycles	13
Planes	12
Trucks and Buses	11
Ships	9
Trains	3

7 rows in set (0.01 sec)

Q2. Find products from different product lines that were ordered together
Show only the top 3 most frequent combinations. Display in descending order according to the total number of orders together.

productLine1	productLine2	ordersTogether
Classic Cars	Vintage Cars	106
Classic Cars	Trucks and Buses	72
Ships	Vintage Cars	63

Q3. Find customers who have never placed an order but have been assigned a sales representative, along with their sales rep's first name and sort by customer name.

Customer	SalesRep	OfficeCity
American Souvenirs Inc	Foon Yue	NYC
Precious Collectables	Martin	Paris

Q4. For each sales representative, calculate the following:

1. Total number of customers they manage
2. Total value of all orders from their customers

Only include sales reps whose customers have placed orders worth more than \$800,000 in total. Sort by the total order value

SalesRep	TotalCustomers	TotalOrderValue
Gerard	7	1258577.81
Leslie	6	1081530.54
Pamela	10	868220.55

Solve Q5 to Q8 using Cross Product. No marks will be given if solved using SET operations, subqueries or Joins.

Q5: Solve Q1

Q6: Solve Q2

Q7: Solve Q3

Q8: Solve Q4